USB-C / magnetic pogo input -> F1 polyfuse -> CN3058E LiFePO4 charger. CC1/CC2 5.1k pull-downs so any USB-C charger gives 5 V.

U4 USBLC6 = ESD on D-/D+/VBUS. J2 is the same signals for an enclosure-wall magnetic connector (fit J1 or J2).

Charger: ICH = 1218 V / R3 = 0.31 A (0.5C). R4 0R: TEMP low = NTC off.

RED = charging, GREEN = full (LEDs run off VBUS only).

Battery: LiFePO4 14500 (2.9-3.6 V). Q1 = reverse protection.

U3 RT9080-33 (2 uA Iq) keeps 3.6 V charge off the C3.

VBAT sense: R8/R9 2.2M/2.2M, always on (0.8 uA).

VBUS_DET (IO5, RTC GPIO) wakes the C3 on charger plug-in.

J1
USB-C 16P

VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
A4 VBUS_IN
A5 CC1
B5 CC2
A7 USB_DN
B7 USB_DN
A6 USB_DP
B6 USB_DP
A8
B8
S1
A1
GND GND

VBUS_IN
F1
0.75A
VBUS

VBUS
C1
10u
GND

VBAT
C2
10u
GND

U2
CN3058E
VBUS 4
VBUS 8
TEMP 1
ISET 2
VIN
CE
TEMP
ISET
GND
BAT
CHRG
DONE
EP
3
9
5 VBAT
7 CHRG_N
6 DONE_N
GND GND

TEMP
R4
0R
GND
ISET
R3
3.9k
GND

CC1
R1
5.1k
GND
CC2
R2
5.1k
GND

CHG_A
R5
1k
VBUS
DONE_A
R6
1k
VBUS

CHRG_N
CHG_A
D2
RED (back)
DONE_N
DONE_A
D3
GREEN (back)

U4
USBLC6-2SC6
VBUS
USB_DN 1
USB_DP 3
5
6 USB_DN
4 USB_DP
2
GND

VBUS_IN 1
USB_DN 2
USB_DP 3
GND 4
J2
MAG-POGO

PWR_FLAG
PWR_FLAG
PWR_FLAG
VBUS
VBAT
GND

VBAT_RAW
BT1
IFR14500 600mAh
GND

Q1_G 1
VBAT_RAW
VBAT 2
Q1
AO3401A
VBAT
C3
47u
GND

Q1_G
R7
100k
GND

U3
RT9080-33GJ5
VBAT 1
VBAT 3
VIN
EN
VOUT
GND
5
2
3.3
GND

+3V3
C4
10u
GND
+3V3
C5
100n
GND

VBAT_ADC
C6
100n
GND

VBAT
R8
2.2M
VBAT_ADC
VBAT_ADC
R9
2.2M
GND

VBUS
R11
100k
VBUS_DET
VBUS_DET
R12
100k
GND

ESP32-C3-WROOM-02. Straps: IO2 (LoRa NSS) + IO8 (LoRa BUSY) pulled up, IO9 = BOOT/PAIR. USB-Serial/JTAG on IO18/IO19 = flash + log.

SHT40 on the BACK side (with SW1, D2, D3). Milled slot around U5 + C12; nothing hot nearby.

+3V3
R13
10k
EN
+3V3
R14
10k
BOOT
+3V3
R15
10k
LORA_NSS
+3V3
R16
10k
LORA_BUSY

EN
C9
1u
BOOT
GND
SW1
PAIR/BOOT (back)
GND

+3V3
C7
10u
GND
+3V3
C8
100n
GND

U1
ESP32-C3-WROOM-02-N4
+3V3
EN 2
XTAL32_P 18
XTAL32_N 17
LORA_NSS 16
VBAT_ADC 15
LORA_DIO1 3
VBUS_DET 4
SDA 5
SCL 6
LORA_BUSY 7
BOOT 8
LORA_SCK 10
USB_DN 13
USB_DP 14
EN
3V3
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO18
IO19
GND
1
9
IO20/RXD 11 LORA_MISO
IO21/TXD 12 LORA_MOSI
GND

+3V3
R18
4.7k
SDA
+3V3
R19
4.7k
SCL

SHT40-AD1B, I2C 0x44.
U5
SHT40-AD1B-R2
SCL
SDA
VDD
VSS
3
1
4
+3V3
SDA
GND
+3V3
C12
100n (back)
GND

Y1 32.768 kHz on IO0/IO1 = RTC slow clock (CONFIG_RTC_CLK_SRC_EXT_CRYS). Keep IO0/IO1 traces short, guard with GND.

XTAL32_P
XTAL32_N
Y1
32.768kHz 12.5pF

XTAL32_P
C10
12p
GND
XTAL32_N
C11
12p
GND

+3V3
C13
10u
GND
+3V3
C14
100n
GND

+3V3
U6
ISC-SX1262-B
VDD
3
LORA_MOSI 14
LORA_MISO 13
LORA_SCK 12
LORA_NSS 15
EN 4
MOSI
MISO
SCK
NSS
RESET
ANT
DIO0/TX_EN
DIO1
DIO2
DIO3
DIO4/BUSY
DIO5/RX_EN
1 RF_ANT
5
6 LORA_DIO1
7
8
10 LORA_BUSY
11
GND
GND
GND
9
16
2
GND GNDGND

AE1
866MHz spring
RF_ANT

LoRa option (TH2-L): U6 SX1262

NSS IO2, SCK IO10, MOSI IO21, MISO IO20

BUSY IO8, DIO1 IO4 (RTC wake), NRESET = EN

TH2-W: leave U6, AE1, C13, C14 off

+3V3
J4
PROG
GND
BOOT 4
USB_DN 5
USB_DP 6

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Agri Sensors and Controls

Sheet: /

File: TH2-FieldNode.kicad_sch

Title: TH2 field node v0.2 — one 40x60 PCB in the common AWD/TH box

Size: A3	Date: 2026-10-09	**Rev: 0.2**
KiCad E.D.A. 9.0.7		Id: 1/1